SEMESTER PROJECT DOCUMENTATION

T.R.A.C.K

Tracking Records And Criminal Knowledge

SUBMITTED TO:

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SYSTEM DESCRIPTION

1. Introduction

Law enforcement agencies deal with a large volume of criminal data on a daily basis. Managing this data manually through paper records or disconnected systems leads to inefficiency, delayed access to records, and difficulty in maintaining consistency between criminal cases, FIRs, victims, and evidence.

TRACK (Tracking Records And Criminal Knowledge) is a database driven crime management system designed to digitize and organize criminal records within a police department. The system enables police officers and administrators to efficiently manage crimes, criminals, cases, FIRs, evidence, and victims using a centralized relational database.

The project is focused on proper database design using ER modeling and normalization techniques up to BCNF to ensure minimum redundancy, data consistency, and integrity.

2. Business Scenario

A district level police station handles multiple criminal reports daily including theft, fraud, assault, cybercrime, and organized criminal activities. Traditionally, these records are maintained manually through registers and physical files. This causes several operational problems:

1. Difficulty in searching historical records.
2. Delayed case tracking and investigation updates.
3. Redundant data entry across departments.
4. Increased risk of data inconsistency and loss.
5. Lack of centralized evidence management.

TRACK replaces the manual process with a relational database management system where officers can register crimes, file FIRs, manage investigations, store evidence, and maintain criminal records in a structured environment. The system also supports role-based access control:

1. Admin Officers can manage offices, officers, and overall records.
2. Regular Officers can manage crime investigations, FIRs, cases, victims, and evidence.

3. System Objectives

1. Maintain a centralized criminal database.
2. Reduce redundancy and inconsistency in records.
3. Support efficient crime investigation tracking.
4. Manage FIRs and criminal records digitally.
5. Store evidence linked to investigation cases.
6. Implement role-based access control.
7. Ensure data integrity using normalization.

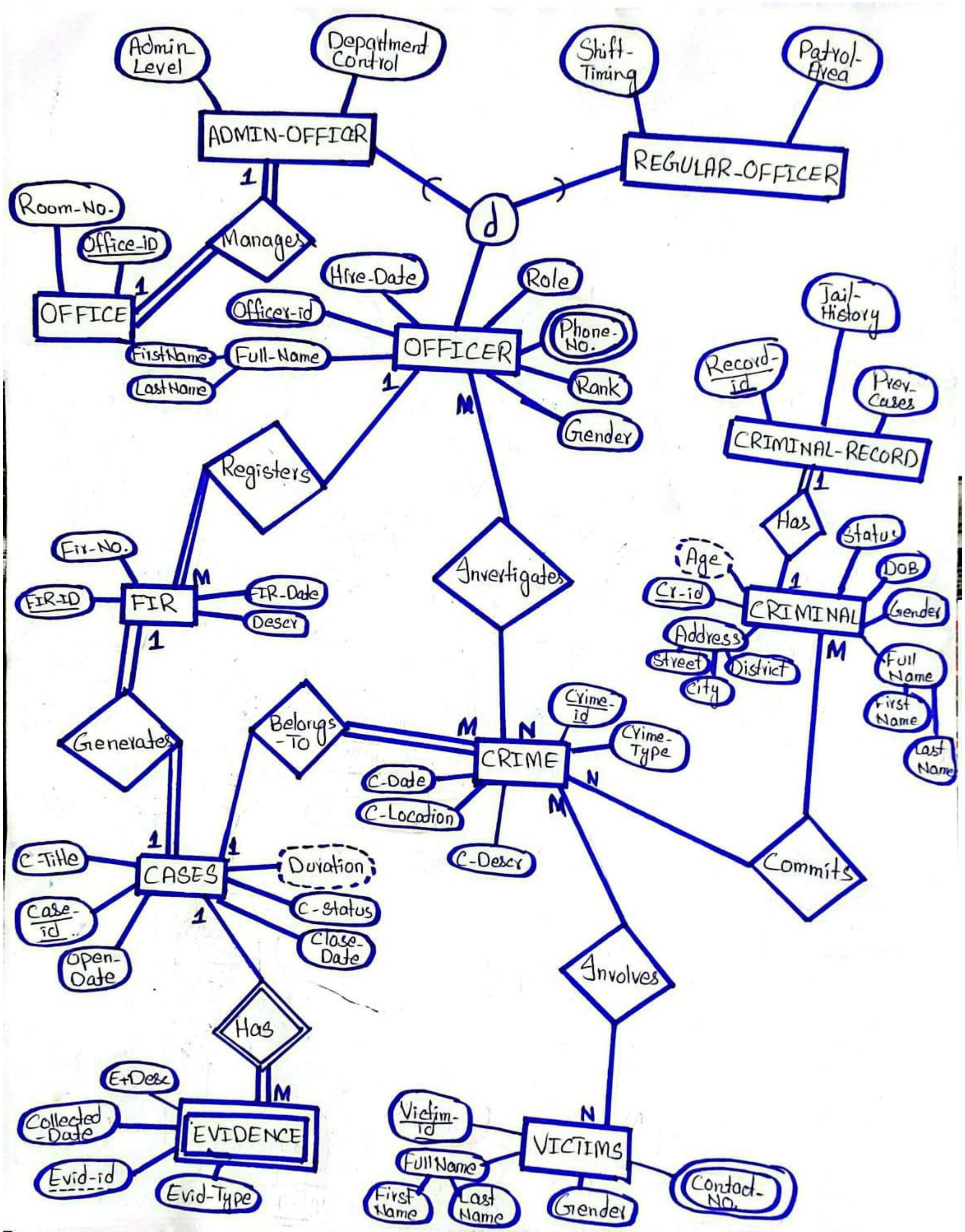
4. Entities Overview

No	Entity	Description
1	OFFICER	Police personnel managing investigations
2	ADMIN_OFFICER	Officers with administrative authority
3	REGULAR_OFFICER	Officers responsible for field operations
4	OFFICE	Police office managed by admin officers
5	FIR	Official First Information Reports
6	CASES	Investigation records linked with FIRs
7	EVIDENCE	Evidence collected during investigations
8	CRIME	Criminal incidents
9	CRIMINAL	Offenders involved in crimes
10	CRIMINAL_RECORD	Historical criminal records
11	VICTIMS	Victims affected by crimes

5. Scope and Limitations

This system is designed for station-level use and does not cover court proceedings, forensic laboratory workflows, or inter-department data sharing. It focuses on the operational needs of a single police station: recording, managing, and retrieving crime-related data through a structured relational database backed by a Java-based graphical user interface connected to an Oracle XE database.

ER DIAGRAM


Admin Level
Department Control
Shift-Timing
Patrol-Area
ADMIN-OFFICER
REGULAR-OFFICER
d
Room-No.
Office-ID
Manages
OFFICE
Hire-Date
Role
Officer-id
Phone-No.
OFFICER
FirstName
Full-Name
LastName
Rank
Gender
Jail-History
Record-id
Prev-Cases
CRIMINAL-RECORD
Has
Status
Age
Registers
Investigates
Fir-No.
FIR-ID
FIR
FIR-Date
Descr
Cr-id
CRIMINAL
DOB
Gender
Address
Street
District
City
Full Name
First Name
Last Name
Crime-id
Crime-Type
Generates
Belongs-TO
CRIME
C-Date
C-Location
C-Descr
Commits
C-Title
CASES
Duration
Case-id
C-status
Close-Date
Open-Date
Involves
Has
E-Desc
EVIDENCE
Collected-Date
Evid-id
Evid-Type
Victim-id
VICTIMS
FullName
First Name
Last Name
Gender
Contact-No.

RELATIONAL SCHEMA (BCNF)

OFFICE(Office-ID PK, Room-No, Officer-id UNIQUE FK)

OFFICER(Officer-id PK, FirstName, LastName, Hire-Date, Role, Rank, Gender)

OFFICER_PHONE(Officer-id FK, Phone-No, PK(Officer-id, Phone-No))

ADMIN_OFFICER(Officer-id PK FK, Admin-Level, Department-Control)

REGULAR_OFFICER(Officer-id PK FK, Shift-Timing, Patrol-Area)

FIR(FIR-ID PK, Fir-No UNIQUE, FIR-Date, Descr, Officer-id FK)

CASES(Case-id PK, C-Title, Open-Date, Close-Date, C-Status, FIR-ID FK)

EVIDENCE(Case-id FK, Evid-id, E-Desc, Collected-Date, Evid-Type, PK(Case-id, Evid-id))

CRIME(Crime-id PK, Crime-Type, C-Date, C-Location, C-Descr)

VICTIMS(Victim-id PK, FirstName, LastName, Gender)

VICTIM_CONTACT(Victim-id FK, Contact-No, PK(Victim-id, Contact-No))

CRIMINAL(Cr-id PK, FirstName, LastName, Street, District, City, Status, DOB, Gender)

CRIMINAL_RECORD(Cr-id PK FK, Record-id UNIQUE, Jail-History, Prev_Cases)

INVESTIGATES(Officer-id FK, Crime-id FK, PK(Officer-id, Crime-id))

BELONGS_TO(Case-id FK, Crime-id FK, PK(Case-id, Crime-id))

INVOLVES(Crime-id FK, Victim-id FK, PK(Crime-id, Victim-id))

COMMITS(Cr-id FK, Crime-id FK, PK(Cr-id, Crime-id))

Structure of Oracle Database Implementation

1. Database Design Approach

The database implementation is based on the final BCNF normalized schema of the TRACK system. The schema minimizes redundancy, enforces referential integrity, and supports efficient relational operations.

2. Oracle Version

Oracle Database XE 21c

Features used:
1. GENERATED AS IDENTITY for automatic primary key generation
2. Foreign key constraints
3. CHECK constraints
4. UNIQUE constraints
5. ON DELETE CASCADE for bridge tables and dependent records

3. Naming Convention

The implementation uses Oracle friendly uppercase naming conventions.

Example:
OFFICER_ID
FIRST_NAME
OPEN_DATE

Reason:
1. Avoids quoted identifiers
2. Easier SQL writing
3. Cleaner Oracle compatibility

4. Automatic Primary Key Generation

All major entity tables use:
GENERATED ALWAYS AS IDENTITY

This automatically generates numeric primary keys.

Example:
OFFICER_ID NUMBER GENERATED ALWAYS AS IDENTITY

5. Composite Keys

Bridge tables and weak entities use composite primary keys.

Examples:
OFFICER_PHONE
VICTIM_CONTACT
INVESTIGATES
BELONGS_TO
INVOLVES
COMMITS
EVIDENCE

6. Referential Integrity

Foreign key constraints enforce relationships between entities.

Examples:
FIR → OFFICER
CASES → FIR
EVIDENCE → CASES
COMMITS → CRIMINAL + CRIME

7. Cascading Deletes

Bridge tables and dependent relations use:
ON DELETE CASCADE

This automatically removes dependent rows when parent records are deleted.

8. CHECK Constraints

CHECK constraints enforce valid domain values.

Examples:
GENDER IN ('Male', 'Female')
CASE_STATUS IN ('Open', 'Closed', 'Under Investigation')

9. Weak Entity Handling

EVIDENCE is implemented as a weak entity using:
PRIMARY KEY (CASE_ID, EVID_ID)

where CASE_ID references the owning CASE.